

Problem 6.4

Unrefereed candidate proof by GPT-6 Astra

Writeup: gpt-6-astra (single model pass)

Catalog id 2604.09449__03 — generated 2026-09-17

UNREFEREED MODEL OUTPUT. This document was selected solely because GPT-6 Astra labelled its own result would_publish: true and returned proved or disproved. It has not passed the adversarial LLM referee used for the earlier Sol campaign, has not been checked by a human mathematician, and has not been checked for novelty. Treat every mathematical and bibliographic claim below as unverified.

Summary and provenance

Catalog id	2604.09449__03
Catalog entry	Problem 6.4
Source corpus	arxiv
Campaign leg	selected second attempts (attacks_retry)
Source paper / entry	Colour-balanced subgraphs
Model verdict	proved (confidence: high)
Model’s one-line claim	The requested $O(k^2)$ bound improves to $O(k \log(k+1))$ for Hamilton cycles in both complete and complete bipartite graphs, including for bounded vector labels in an arbitrary norm.
Model	gpt-6-astra, reasoning effort max, mode=pro, flex service tier
Original artifact	attacks_retry/2604.09449__03/output.md
Independent review	None
Model’s caveats	Optimality and priority relative to later revisions of the source paper are not established.

Problem statement

No extracted statement was stored in the artifact metadata.

Astra writeup

The text below is the model output verbatim, apart from moving its machine-readable verdict block into the summary above and shifting Markdown heading levels for this document. Mechanical TeX defects and equation tags were normalized where needed for compilation.

An $O(k \log k)$ bound

The earlier attempt’s midpoint-splicing and approximate-convexity arguments are sound. Below I verify both, sharpen the midpoint estimate, and extend the complete-graph argument to arbitrary bounded vector labels. Thus the improvement does not depend on interpreting f_h only as a recolouring statistic.

Here “proved” refers to the requested improvement in the upper bound, not to determining its optimal order.

1. Precise statement

For $d \geq 1$, put

$$B_d = \lceil \log_2 d \rceil + 2.$$

If h labels the edges of a graph by vectors, write

$$h(F) = \sum_{e \in E(F)} h(e).$$

Theorem 1 Let X be a real d -dimensional normed space, and suppose that

$$h : E(G) \longrightarrow X, \quad \|h(e)\| \leq L.$$

1. If $G = K_{n,n}$, where $n \geq 2$, then G has a Hamilton cycle H satisfying

$$\left\| h(H) - \frac{2}{n} h(G) \right\| \leq 6LdB_d. \quad (1)$$

2. If $G = K_N$, where $N \geq 3$, then G has a Hamilton cycle H satisfying

$$\left\| h(H) - \frac{2}{N-1} h(G) \right\| \leq 6LdB_d + 8Ld + 2L. \quad (2)$$

The centring vectors in (1) and (2) are the expected label sums of uniformly random Hamilton cycles. In particular, for unit-bounded, zero-sum d -dimensional labels, the least norm of a Hamilton-cycle label sum is $O(d \log(d+1))$.

For the recolouring formulation, the theorem gives the following explicit bounds.

Corollary 2 Let $k \geq 2$, and suppose that the host graph is k -edge-coloured with equally large colour classes.

- If the host is $K_{n,n}$, $n \geq 2$, and $k \mid 2n$, some Hamilton cycle can be made colour-balanced by recolouring at most

$$3kB_k \quad (3)$$

of its edges.

- If the host is K_N , $N \geq 3$, and $k \mid N$, some Hamilton cycle can be made colour-balanced by recolouring at most

$$3kB_k + 4k + 1 \quad (4)$$

of its edges.

Consequently, the quadratic upper bound in the question is replaced by

$$\boxed{O(k \log(k+1))}.$$

The substantive ingredient is an improved vector-labelled bipartite matching bound.

2. Halving a vector-valued necklace

We first establish the continuous splitting fact needed in the matching argument.

Lemma 3 Let $f : [0, r] \rightarrow \mathbb{R}^d$ be integrable. There is a set $A \subseteq [0, r]$ whose indicator is constant between at most d cut points and such that

$$\int_A f(x) dx = \frac{1}{2} \int_0^r f(x) dx. \quad (5)$$

Proof For $x = (x_0, \dots, x_d) \in S^d$, partition $[0, r]$ into consecutive intervals $J_0, \dots, J_d$ of lengths

$$rx_0^2, \dots, rx_d^2.$$

Define

$$F(x) = \sum_{i=0}^d \operatorname{sgn}(x_i) \int_{J_i} f(t) dt.$$

The map $F : S^d \rightarrow \mathbb{R}^d$ is continuous: when a coordinate tends to zero, its interval has vanishing length, so the corresponding integral tends to zero. It is also odd.

By the Borsuk–Ulam theorem, F has a zero. Taking A to be the union of the positively signed intervals gives (5). There are at most d internal cut points. $\square$

No positivity assumption on the coordinates of f is needed.

3. Approximate midpoint closure for matching sums

Lemma 4 Suppose that the edges of $K_{r,r}$ are labelled by vectors in X , each of norm at most L . For any two perfect matchings P, Q , there is a perfect matching R such that

$$\left\| h(R) - \frac{h(P) + h(Q)}{2} \right\| \leq 3Ld. \quad (6)$$

Proof Let the bipartition be $U \cup V$. Regard P, Q as bijections from V to U , and define a permutation σ of V by

$$Q(v) = P(\sigma(v)).$$

Order the vertices of V by concatenating cyclic orders of the cycles of σ . Write the resulting order as $v_1, \dots, v_r$. Let P_i, Q_i denote the respective matching edges incident with v_i .

On $I_i = [i-1, i]$, put

$$f(x) = h(P_i) - h(Q_i).$$

Apply Lemma 3, after identifying X linearly with $\mathbb{R}^d$, and set

$$\alpha_i = |A \cap I_i|.$$

Then

$$\sum_i \alpha_i (h(P_i) - h(Q_i)) = \frac{1}{2} \sum_i (h(P_i) - h(Q_i)). \quad (7)$$

Choose $s_i \in \{0, 1\}$ nearest to α_i , breaking ties arbitrarily. At most d of the numbers α_i lie strictly between zero and one, since each such unit interval contains an internal cut point. Therefore

$$\sum_i |s_i - \alpha_i| \leq \frac{d}{2}. \quad (8)$$

Select P_i if $s_i = 1$, and Q_i if $s_i = 0$. This gives one selected edge at every vertex of V , though it need not yet be a matching. Let z be its label sum. By (7) and (8),

$$\begin{aligned} \left\| z - \frac{h(P) + h(Q)}{2} \right\| &= \left\| \sum_i (s_i - \alpha_i) (h(P_i) - h(Q_i)) \right\| \\ &\leq 2L \sum_i |s_i - \alpha_i| \leq Ld. \end{aligned} \quad (9)$$

We now bound the number of matching defects.

For each i , the chosen value s_i occurs as the indicator of A at some non-cut point in the interior of I_i . Choosing such points in increasing order shows that the linear binary sequence

$$s_1, \dots, s_r$$

has at most d changes.

Consider one cycle block of σ , and let c be its number of internal linear changes. Closing that block cyclically adds at most one change; if it adds one, then $c \geq 1$. Hence the number of cyclic changes in that block is at most $2c$. Summing over all blocks, the total number b of cyclic changes satisfies

$$b \leq 2d. \quad (10)$$

For a cycle written so that $\sigma(v_i) = v_{i+1}$, the degree of $P(v_i)$ in the selected edge set is

$$1 + s_i - s_{i-1},$$

with cyclic subscripts. Consequently, there are exactly $b/2$ vertices of U of degree two and $b/2$ of degree zero.

Delete one selected edge at each degree-two vertex. This frees $b/2$ vertices of V . Match them arbitrarily to the degree-zero vertices of U , using completeness of the host. The resulting graph is a perfect matching R , obtained by replacing $b/2 \leq d$ selected edges. Thus

$$\|h(R) - z\| \leq 2L \frac{b}{2} \leq 2Ld. \quad (11)$$

Combining (9) and (11) proves (6). $\square$

The important feature is that the error is independent of r .

4. Approximate convexification costs only a logarithm

Lemma 5 Let S be a finite subset of a d -dimensional normed space. Suppose that for every $u, v \in S$, some $w \in S$ satisfies

$$\left\| w - \frac{u+v}{2} \right\| \leq D. \quad (12)$$

Then every $x \in \text{conv}(S)$ satisfies

$$\text{dist}(x, S) \leq DB_d. \quad (13)$$

Proof By Carathéodory's theorem, write

$$x = \sum_{i=1}^m \beta_i x_i, \quad x_i \in S, \quad m \leq d+1.$$

The case $m = 1$ is immediate.

First suppose that all coefficients have a common dyadic denominator 2^q . Form a complete binary tree with 2^q leaves, placing the required copies of $x_1, \dots, x_m$ in consecutive blocks.

Assign representatives in S to the internal nodes, proceeding upwards. If all leaves below a node are equal, choose that common point. Otherwise use (12) on its two child representatives.

At depth j from the root, the number H_j of nodes whose leaves are not all equal is at most

$$H_j \leq \min(2^j, m-1). \quad (14)$$

Indeed, their disjoint leaf intervals must each contain a boundary between consecutive blocks.

An error introduced at depth j contributes to the root with coefficient 2^{-j} . Therefore the root representative $z \in S$ satisfies

$$\begin{aligned} \|z - x\| &\leq D \sum_{j=0}^{q-1} 2^{-j} H_j \\ &\leq D \sum_{j \geq 0} \min(1, (m-1)2^{-j}) \quad (15) \\ &\leq D(\lceil \log_2(m-1) \rceil + 2) \\ &\leq DB_d. \end{aligned}$$

For arbitrary coefficients, approximate them by dyadic coefficients summing to one. Since S is finite, a subsequence of the resulting root representatives is constant. Passing to the limit proves (13). $\square$

This is where the quadratic dependence is avoided: a sequential convexification can pay an $O(d)$ error $O(d)$ times, whereas the weighted dyadic construction pays only $O(\log d)$ levels.

Proposition 6: vector matching bound For an edge-labelling of $K_{r,r}$ in X with $\|h(e)\| \leq L$, there is a perfect matching M such that

$$\left\| h(M) - \frac{1}{r} h(K_{r,r}) \right\| \leq 3LdB_d. \quad (16)$$

In fact, every point in the convex hull of the perfect-matching label sums is within $3LdB_d$ of a perfect-matching label sum.

Proof Apply Lemma 5 to

$$S = \{h(M) : M \text{ is a perfect matching}\},$$

using Lemma 4 with $D = 3Ld$.

A uniformly random perfect matching contains each edge with probability $1/r$, so

$$\frac{1}{r} h(K_{r,r}) \in \text{conv}(S).$$

This proves (16). $\square$

5. Hamilton cycles in complete bipartite graphs

Let the parts of $K_{n,n}$ be

$$A = \{a_1, \dots, a_n\}, \quad B,$$

with A cyclically ordered and $a_{n+1} = a_1$.

Construct an auxiliary $K_{n,n}$ between the gaps

$$g_i = (a_i, a_{i+1})$$

and the vertices of B . Give its edges the labels

$$\lambda(g_i b) = h(a_i b) + h(a_{i+1} b). \quad (17)$$

These labels have norm at most $2L$.

A perfect matching in the auxiliary graph assigns distinct vertices b_i to the gaps and produces the Hamilton cycle

$$a_1 b_1 a_2 b_2 \cdots a_n b_n a_1. \quad (18)$$

Its h -sum is exactly the auxiliary matching's λ -sum.

Each original edge appears in exactly two auxiliary labels. Hence

$$\frac{1}{n} \sum_{i,b} \lambda(g_i b) = \frac{2}{n} h(K_{n,n}). \quad (19)$$

Applying Proposition 6 with label bound $2L$ gives (1).

This construction also works for $n = 2$; it produces the unique four-cycle.

6. Hamilton cycles in complete graphs

We need a nearly balanced cut, now for arbitrary normed vector labels.

Lemma 7 Let $N \geq 4$, and suppose the edges of K_N have labels in X of norm at most L . Put

$$a = \left\lceil \frac{N}{2} \right\rceil, \quad b = \left\lfloor \frac{N}{2} \right\rfloor, \quad p = \frac{2ab}{N(N-1)}.$$

There is a partition $V(K_N) = A \cup B$, with $|A| = a$, $|B| = b$, such that

$$\|h(E(A, B)) - p h(K_N)\| \leq 2LdN. \quad (20)$$

Proof Choose A uniformly among the a -subsets, and let I_e indicate that e crosses the partition. Thus $\mathbb{E} I_e = p$.

We first record a scalar variance bound. Let $|w_e| \leq L$, and put

$$Z = \sum_e w_e (I_e - p).$$

For adjacent distinct edges e, f ,

$$\mathbb{P}(I_e = I_f = 1) = \frac{p}{2},$$

and therefore

$$|\text{Cov}(I_e, I_f)| = \left| p \left(\frac{1}{2} - p \right) \right| \leq \frac{1}{N}. \quad (21)$$

For disjoint edges, direct enumeration gives

$$\text{Cov}(I_e, I_f) = \begin{cases} \frac{N}{2(N-1)^2(N-3)}, & N \text{ even}, \\ \frac{N+1}{2N^2(N-2)}, & N \text{ odd}. \end{cases} \quad (22)$$

Both are at most $4/N^2$. For example, the joint crossing probability before subtracting p^2 is

$$\frac{4a(a-1)b(b-1)}{N(N-1)(N-2)(N-3)}.$$

There are $\binom{N}{2}$ edges, $3\binom{N}{3}$ unordered adjacent edge pairs, and $3\binom{N}{4}$ unordered disjoint edge pairs. Using absolute values for the covariance terms,

$$\begin{aligned} \text{Var } Z &\leq L^2 \left(\frac{1}{4} \binom{N}{2} + \frac{2}{N} 3 \binom{N}{3} + \frac{8}{N^2} 3 \binom{N}{4} \right) \\ &\leq \frac{17}{8} L^2 N^2 < 4L^2 N^2. \end{aligned} \quad (23)$$

Choose a basis $u_1, \dots, u_d$ of X with

$$\|u_j\| = 1$$

and coordinate functionals φ_j of dual norm one. Such a basis exists: maximize the absolute determinant of d vectors in the unit ball. Replacing a basis vector by any other unit-ball vector shows that every corresponding coordinate has absolute value at most one. Also,

$$\|x\| \leq \sum_{j=1}^d |\varphi_j(x)|. \quad (24)$$

Set

$$T = \sum_e (I_e - p)h(e).$$

Applying (23) to the scalar weights $\varphi_j(h(e))$, which have absolute value at most L , gives

$$\begin{aligned} \mathbb{E}\|T\| &\leq \sum_{j=1}^d \mathbb{E}|\varphi_j(T)| \\ &\leq \sum_{j=1}^d \sqrt{\mathbb{E}[\varphi_j(T)^2]} \\ &\leq 2LdN. \end{aligned}$$

Some partition therefore satisfies (20). $\square$

Completing the proof of Theorem 1 For $N = 3$, the Hamilton cycle consists of all three edges, and the expression on the left of (2) is zero. Assume henceforth that $N \geq 4$.

Choose A, B as in Lemma 7, cyclically order

$$A = \{a_1, \dots, a_a\},$$

and let

$$T = h(E(A, B)), \quad S = h(K_N).$$

If N is even, $|B| = a$. If N is odd, adjoin a dummy symbol $*$ to B , obtaining a set B^* of size a .

Construct an auxiliary $K_{a,a}$ between the cyclic gaps $g_i = (a_i, a_{i+1})$ and B^* . For a real vertex $v \in B$, set

$$\lambda(g_i v) = h(a_i v) + h(a_{i+1} v). \quad (25)$$

In the odd case, set

$$\lambda(g_i *) = h(a_i a_{i+1}). \quad (26)$$

All auxiliary labels have norm at most $2L$.

Every auxiliary perfect matching produces a Hamilton cycle: insert a real vertex into its assigned gap, and in the odd case use the direct edge across the dummy-assigned gap.

Let μ be the mean auxiliary perfect-matching label sum. Then

$$\mu = \begin{cases} \frac{2}{a}T, & N \text{ even,} \\ \frac{2}{a}T + \frac{1}{a}Y, & N \text{ odd,} \end{cases} \quad (27)$$

where, in the odd case,

$$Y = \sum_{i=1}^a h(a_i a_{i+1}).$$

If N is even, then

$$\frac{2p}{a} = \frac{2}{N-1},$$

so

$$\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS). \quad (28)$$

If N is odd, then $2p/a = 2/N$, and consequently

$$\mu - \frac{2}{N-1}S = \frac{2}{a}(T - pS) + \frac{Y}{a} - \frac{S}{\binom{N}{2}}. \quad (29)$$

Both Y/a and $S/\binom{N}{2}$ have norm at most L . Using Lemma 7 and $a \geq N/2$, we obtain in either case

$$\left\| \mu - \frac{2}{N-1}S \right\| \leq 8Ld + 2L. \quad (30)$$

By Proposition 6, an auxiliary perfect matching has label sum within

$$6LdB_d$$

of μ . Its corresponding Hamilton cycle therefore satisfies (2). This completes the proof of Theorem 1. $\square$

7. Translation to colour balance

Give every edge of colour j the label

$$h(e) = e_j \in (\mathbb{R}^k, \|\cdot\|_1).$$

Thus every edge label has norm one, and $h(F)$ is the colour-count vector of F .

For a globally colour-balanced $K_{n,n}$,

$$\frac{2}{n}h(K_{n,n}) = \frac{2n}{k}\mathbf{1}.$$

For a globally colour-balanced K_N ,

$$\frac{2}{N-1}h(K_N) = \frac{N}{k}\mathbf{1}.$$

If a subgraph F has kq edges and colour counts $m_1, \dots, m_k$, its minimum recolouring number is

$$\rho(F) = \sum_{j=1}^k (m_j - q)_+ = \frac{1}{2} \sum_{j=1}^k |m_j - q|. \quad (31)$$

Indeed, every surplus edge must be recoloured, and recolouring surplus edges into deficient colours attains this number.

Taking $d = k$, $L = 1$, and halving the bounds in Theorem 1 proves (3) and (4).

Conclusion and scope

The proof establishes, uniformly in the host order,

$$f_h(H) = O(k \log(k+1))$$

in the bounded, zero-sum vector-label formulation, as well as the explicit recolouring bounds above. It also establishes the corresponding $O(k \log(k+1))$ bipartite perfect-matching estimate.

Thus it answers the stated request for an improvement over $O(k^2)$. It does **not** establish an $O(k)$ bound or determine the optimal order. I have not independently checked whether a later revision of the source already contains this improvement.